

Indices

Worksheet 1 (Non-Calculator)

1. W17/qp12/q12

$$a^x = 5$$

a. Find a^{2x} [1]

b. Find a^{-x} [1]

2. S18/qp11/q16

a. Evaluate $3^3 - 3^0$ [1]

b. Simplify completely [2]

$$\left(\frac{9a^3b^3}{16ba^5}\right)^{\frac{1}{2}}$$

3. W18/qp11/q14

a. Evaluate [1]

$$\left(\frac{1}{3}\right)^0 - \left(\frac{1}{3}\right)^2$$

b. Simplify [2]

$$\left(\frac{27}{x^6}\right)^{-\frac{1}{3}}$$

4. W18/qp12/q21

a. Evaluate $9^1 + 9^0$ [1]

b. Find n , where $4^n = 2^{n-1}$ [2]

5. S19/qp11/q6

a. Evaluate.

i. $36^{1/2}$ [1]

ii. 5^{-2} [1]

b. $2^2 \times 8^{\frac{5}{3}} = 2^k$ find the value of k [2]

6. S19/qp12/q20

Simplify [2]

$$\left(\frac{9x^7y}{x^5y^9}\right)^{-\frac{1}{2}}$$

7. W19/qp11/q12

a. Evaluate $3^{-2} \times 3^4$ [1]

b. Evaluate 3×3^0 [1]

c. Simplify $y^{\frac{1}{2}} \times 4y^{\frac{1}{4}}$ [1]

8. W19/qp12/q13

Simplify

a. $(2x^2)^0$ [1]

b. $(3x^3)^2$ [1]

c. $\left(\frac{8}{x^3}\right)^{\frac{1}{3}}$ [2]

9. S20/qp12/q21

Simplify [2]

$$\left(\frac{2x^2}{x^5}\right)^{-3}$$

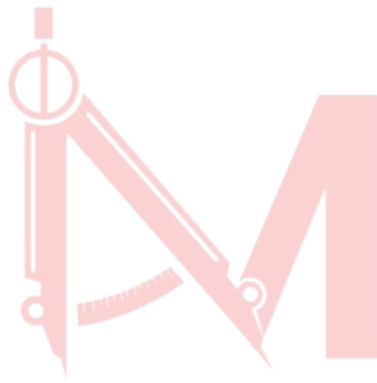
10. W20/qp12/q12

a. Simplify $(2x^2)^3$ [1]

b. Simplify $6t^3 \div \left(\frac{2}{3}t^2\right)$ [2]

Answer Key:

1. (a) 25 (b) $\frac{1}{5}$
2. (a) 26 (b) $\frac{3b}{4a}$
3. (a) $\frac{8}{9}$ (b) $x^{2/3}$
4. (a) 10 (b) -1
5. (a)(i) 6 (a)(ii) $\frac{1}{25}$ (b) 7
6. $(\frac{y^4}{3x})$
7. (a) 9 (b) 2 (c) $4y^{3/4}$
8. (a) 1 (b) $9x^6$ (c) $\frac{x}{2}$
9. $\frac{x^9}{8}$
10. (a) $8x^6$ (b) $9t$



Mathlete
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